COMPLETE FEATURE ATLAS

EVERY PART OF THE KINGDOM, MODULE BY MODULE

Companion to the tri-fold pamphlet. Wellness tracker, not a medical device: estimates are labelled and nothing here diagnoses.

WATER & NUTRITION — PROVISIONS HALL

- **Water Quest:** a layered valley scene that follows the time of day; the knight walks, drinks, carries and pours; well stages and earlier discoveries.
- **Water goal** from body weight (35 mL/kg, the team's own rule), with extras at higher activity levels.
- **Elemental Thermometer:** the water potion adjusts to opt-in live weather (Open-Meteo).
- **Nutrition and hydration on one page;** the Kolej MARA Kulim canteen menu with calories; pick several foods and **LOG SELECTED** once.
- **Add More Food:** Luke searches the web (Gemini with Google Search) for any food, portion and change, with a source for each part and an Atwater energy check. Signed out, the canteen menu.
- **Caloric Ledger** and **Metabolic Forge** in Alchemy.
- **Alchemist's Energy Workshop:** food energy, work against gravity and metabolic use are three different quantities; it shows why.
- **Fruit & veg servings** (80 g portions, WHO).

SLEEP — DREAM REALM

- Sleep duration against recommended ranges (National Sleep Foundation); **Sleep Score** (the team's own game formula, labelled).
- **Log last night's sleep** in one tap, or for any earlier date.
- **Dream Battle:** continuous sleep becomes strength in a torch-lit dungeon with a princess and a deterministic battle.
- **Counting Sheep:** a full-screen animated pixel dream.
- **Wake-up alarm:** in-app alarm with sounds, snooze and a bedside clock; Android Clock intent; iPhone steps; .ics reminder.
- A sleep track of its own in the music.

STRESS & CALM — MIND FOREST

- **DASS-21 check-in:** 21 statements (0–3, past week), three scale scores (× 2, 0–42), shown without severity labels.
- **Storm level** from the total; **Storm Within:** the knight's face, posture and surroundings follow your rating.
- **Calming Grove:** follow a light over the pond as it grows and shrinks with your breath.
- **Relief collection:** Shake It Off, Empower Yourself, Arrow Focus, Reaction Focus, Calming Sounds and the original tools.
- **Wizard's Counsel** scene and breathing work signed out; **crisis help numbers** are shown to everyone.

BODY & TRAINING — ENERGY FORGE

- **BMI** (adult screening), healthy weight range, **BMR** (Mifflin–St Jeor), **TDEE** (FAO/WHO/UNU activity levels), protein, carbohydrate, fat and fibre goals.
- **"Anything that affects your activity?"** question: quest order, stair goals behind a doctor's note and pace notes follow your answers.
- **Training Hall:** goal choice, 22 moves, 19 muscle maps, tutorial, session; Calisthenics is bodyweight only; dizzy and chest-pain stop buttons.
- **Running Road:** GPS tracker for distance, moving time, pace and calories; **Runners' Board** (opt-in).
- **Haze and heat:** live Kulim weather and haze (jerebu) check and a heat tip above Running.

STAIRS & HEART — STAIR MOUNTAIN · HEARTSTONE HALL

- **Casual stairs** (GPS check-in first, then by hand) and **workout with heart rate** (BPM before, after and 1 minute after).
- **Pulse as periodic motion:** f, T and ω; energy and VO_2 by the Group 14 method, behind a "?"; shown as not valid when the pulse after is below the pulse before.
- **Pulse Wave Lab:** camera pulse (photoplethysmography) with live wave and BPM; video never stored.
- **Smartwatch or chest strap** over Web Bluetooth, plus a "Will my watch work?" sheet.
- **Recovery** (1-minute drop), resting-pulse hint, peak heart rate (% of 220 − age, with Tanaka beside it).
- **Stair Prescription:** a weekly stair plan you choose (goal, days, minutes); not a prescription.
- **Group 14 measured stairways** table (rise, run, hypotenuse, angle).

STAIR SCIENCE — DETAILS OF A STAIRCASE

- **Stair Sense (Gravity Trial):** START, PAUSE, FINISH; steps and cadence from the phone's motion sensor; h = n × r, W = mgh, P = W ÷ t.
- **Stair Sight:** 2D step-edge finding in the live camera picture; 3D WebXR (Android Chrome with ARCore) measures rise and tread and builds the stairway; the ghost climbs it.
- **Enchanted Tower, Spectral Pacer, Cadence Echo** (microphone), **Luna's Wind Spell** and **Gravity Anomaly.**
- **Dueling Arena:** race a replay of your own earlier climb (no XP).
- **Ascent Monument:** an endless Tower of Babel from your climbs, one stone per 0.25 m, landmark comparisons, fun facts and a ledger.

COMPLETE FEATURE ATLAS

Continued: the map, the game, the guides, the evidence and the platform

CAMPUS HEALTH OBSERVATORY — THE MAP

- 3D **MapLibre** map of Kolej MARA Kulim from OpenStreetMap, in **Realistic** and **Fantasy** looks, with clustering markers, search and filters.
- **Staircase stops:** a spinning heart on a post in its difficulty colour (Mild, Moderate, Vigorous); solid 3D stops with depth and glow.
- **Game-style HUD:** the HealthWiz button menu, search, Legend, a draggable Details sheet (Stairs and Workout tabs), **NEAREST STAIRS** wave with GPS check-in.
- **Avatars** (Tom, anime school boy and girl); a labelled demo walk away from campus; position never saved or sent.
- **Terrain and ground:** illustrative three-level terrain with contours; Blender buildings coloured from the drone video; roads, kerbs, pitches and car parks.
- **Sky** from live weather, haze and the clock, with a preview button.
- **17 landmarks** with discovery animation, **Map journal**, **Layers** panel, sigils for recorded sessions (no XP).
- **Astral Navigator:** the Sun's position (NOAA method) in a compass.
- **3D stair preview** (Three.js) built from survey measurements, labelled illustrative when not surveyed.
- **Survey importer:** fill a Word sheet, run one command, refresh.

KINGDOM & GAME — BALANCE TOWER

- **Title screen:** time of day, live weather, returning-player ribbon and tap reactions; a loading screen fills the HealthWiz heart pixel by pixel.
- **Kingdom map** of eight regions in five states up to Flourishing, world progression, a chronicle and the Shadow Keep.
- **Quests:** adaptive daily focus quest, weekly quests, Quest Board; **Hall of Heroes** leaderboard (opt-in, game progress only).
- **Fair XP** with a daily allowance and exploration reward; levels with a flowing mana bar; **audited badges; gentle streaks** with rest days and a week view.
- **Arcana:** the Oracle's crystal ball and the **Guild War of Joules** (opt-in).
- **Music:** a recorded home theme plus tracks for Sleep, Water, Stress, Counsel and the Registry; one switch and volume.
- **Traveller's Registry** onboarding; particle effects for level-ups and badges.

GUIDES — MEDIUS · LUNA · LUKE

- **Medius** keeps the app-wide tour and the Wizard's Counsel; he reacts to events through speech bubbles with cooldowns.
- **Luna:** 12 expression portraits and 21 poses, **Meet Luna** in Settings, **Talk with Luna** (AI) and **Ask Luna** (finds a feature or opens a "?", no AI).
- **Luke:** a small round help button on every Health Hall section opens a spotlight walk-through; he also runs Add More Food.
- **Medius notices:** insights from your own baselines and trends, worded as observations, never diagnoses.

INSIGHT & REPORTS — HEALTH HALL

- **Health Score** (five weighted indicators into 0–100) and the **Health Orb** with ranked quest cards.
- **Healer's What-if:** try a gentle change; nothing is saved.
- **Health Time Machine:** day, week or month timeline of what was really saved.
- **Statistics:** touch, keyboard and screen-reader charts; week against week.
- **Weekly Chronicle:** a printable report that goes nowhere unless you save it.
- **Log for…:** yesterday or any earlier date, never the future; **All entries** grouped by day with totals and filters.

SCIENCE & EVIDENCE

- **"?" explanations:** inputs, formula, your own numbers step by step, sources and limits.
- **Evidence Vault:** every source on shelves, with where you see it; calculation cards open from a "?".
- **Science Laboratory:** Be Your Own Experiment, the **Evidence Trial** and **Royal Surveyors** (Truth-Seeker badge).
- **Uncertainty Engine:** how unmeasured assumptions move an estimate; no made-up error margins.
- **Group 14 F21 study:** raw measurements with every report table worked out from them and tested.

ACCOUNT, DATA & PLATFORM

- **Install and offline:** installable app with a service worker; also a single shareable HTML file.
- **Backup and restore;** versioned data with step-by-step upgrades, damaged-data recovery and automatic safety copies.
- **Optional account:** email magic link or code, Google sign-in; cloud sync with offline queue, and a choice on first sign-in: keep device, keep cloud or merge.
- **Privacy:** Row Level Security, delete account and cloud data, AI keys only on the server.
- **Look and comfort:** 8 themes plus custom palettes, High/Balanced/Performance modes, reduced motion, pixel-art emoji, keyboard-friendly dialogs and an error card instead of a blank screen.
- **Quality:** 102 browser test files, performance budgets and a device matrix.